# Bio334 – Building Maximum Likelihood Trees

Solutions to the exercises

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# Location of today's exercises

Within the cloned github repository:

Bio334/05_maximum_likelihood_phylogenies/exercises

# Exercise 1 – Part 1: The grammar

**Is the tree specified as rooted or unrooted?**

((A,(B,(C,D))),(E,F));

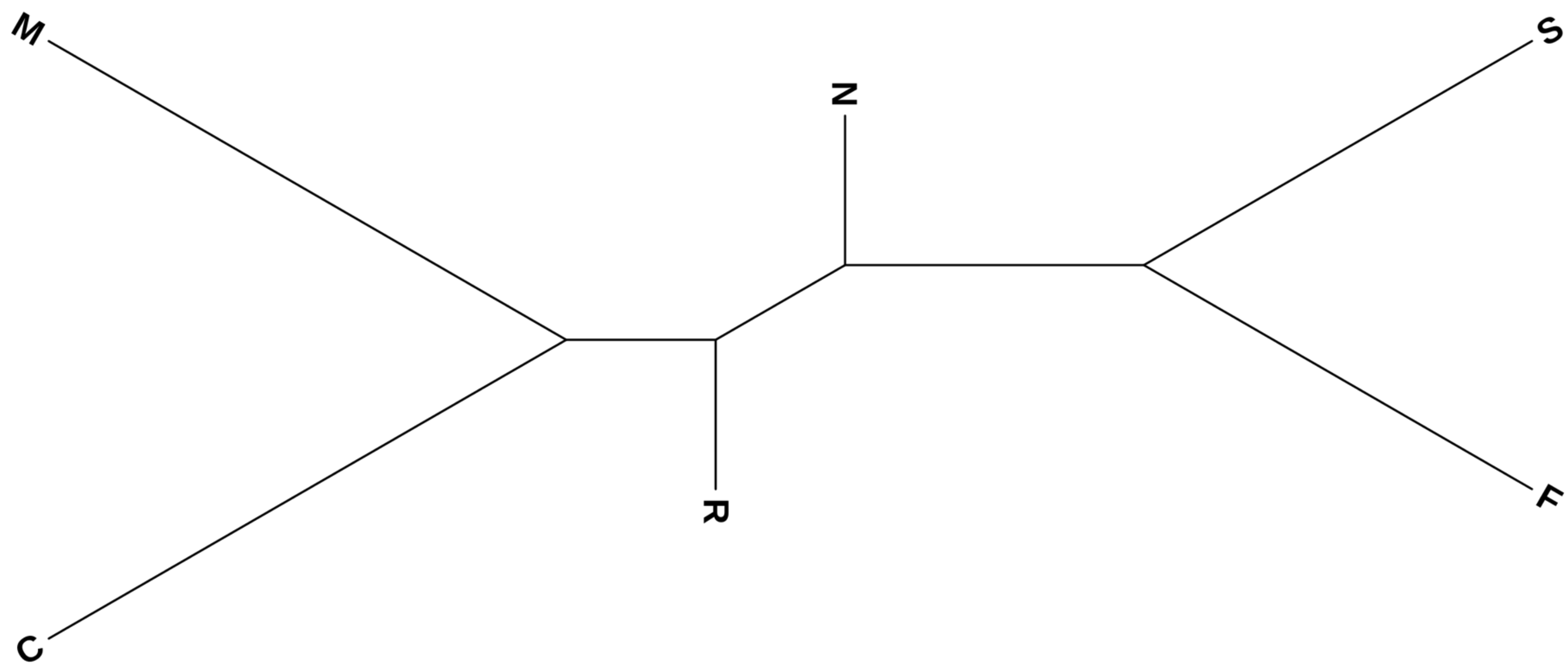
# Exercise 1 – Part 1: The grammar

**Is the tree specified as rooted or unrooted?**

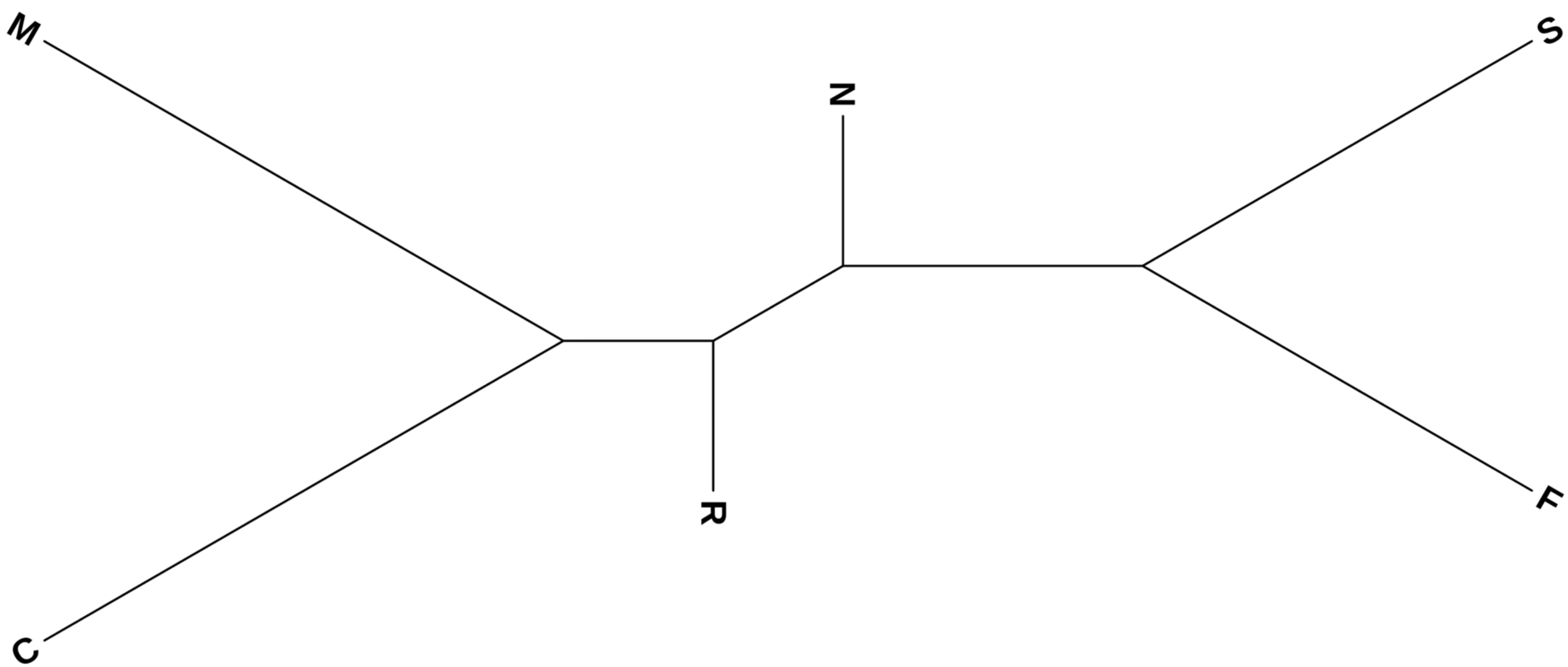
((A,(B,(C,D))),(E,F));

-> The tree is rooted

# Exercise 1 – Part 2: Specifying branch length

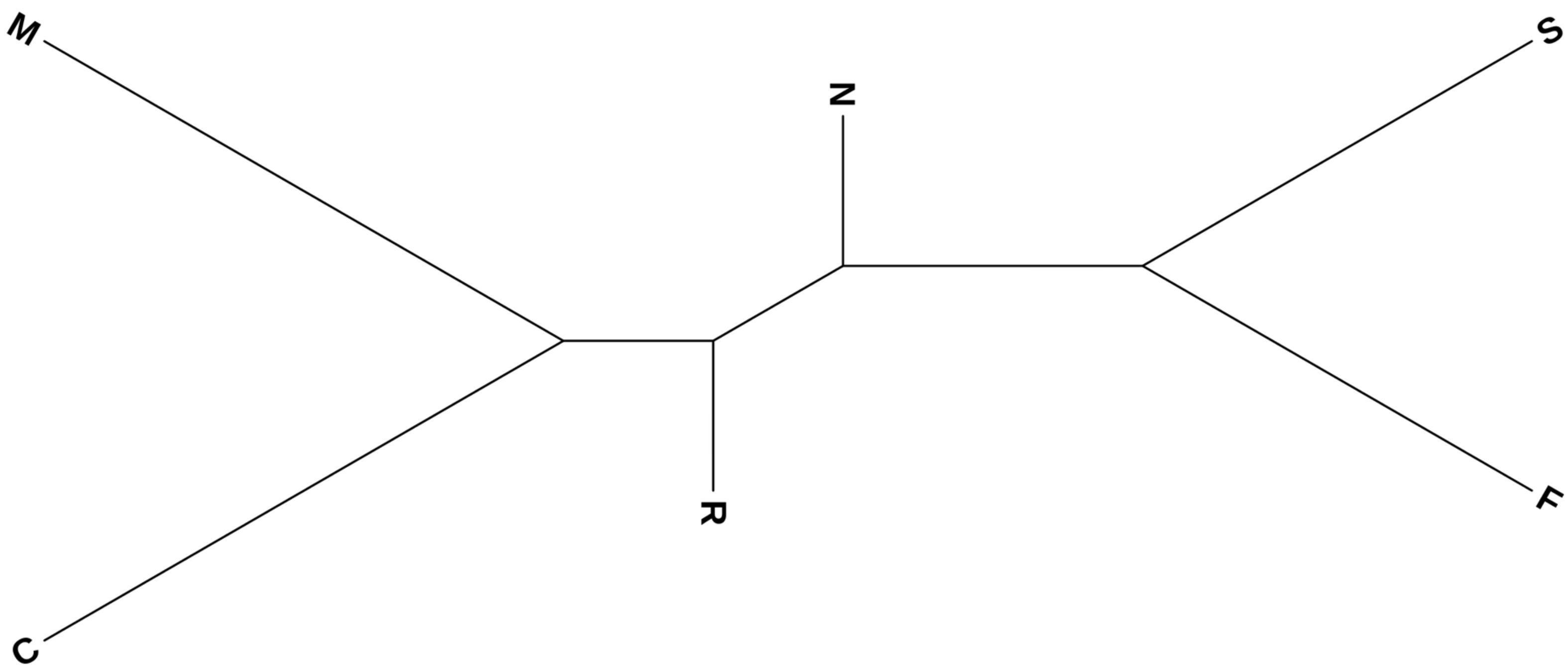


# Exercise 1 – Part 2: Specifying branch length




(N,(R,(C,M)),(S,F));

# Exercise 1 – Part 2: Specifying branch length



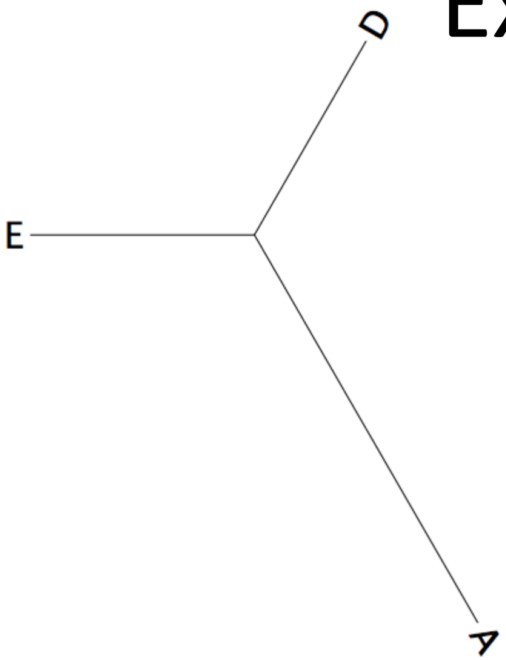

(N:0.5,(R:0.5,(C:2,M:2):0.5):0.5,(S:1.5,F:1.5):1);

# Exercise 1 – Part 3: Common errors in Newick representations

# Exercise 1 – Part 3.a

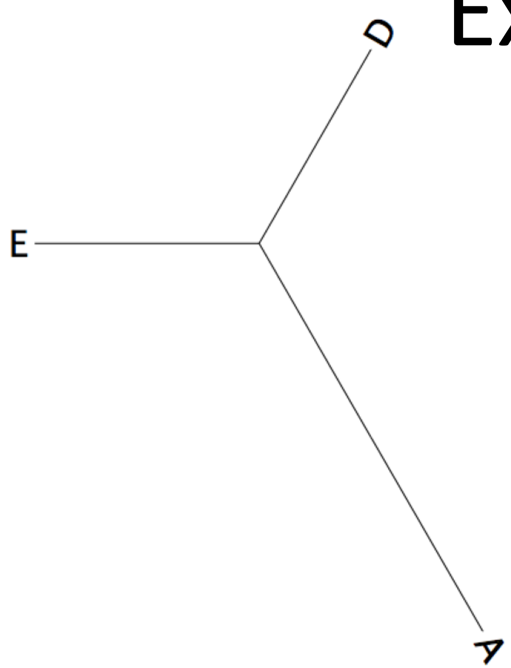
(A,(E,D)),(C,(B,F));

# Exercise 1 – Part 3.a



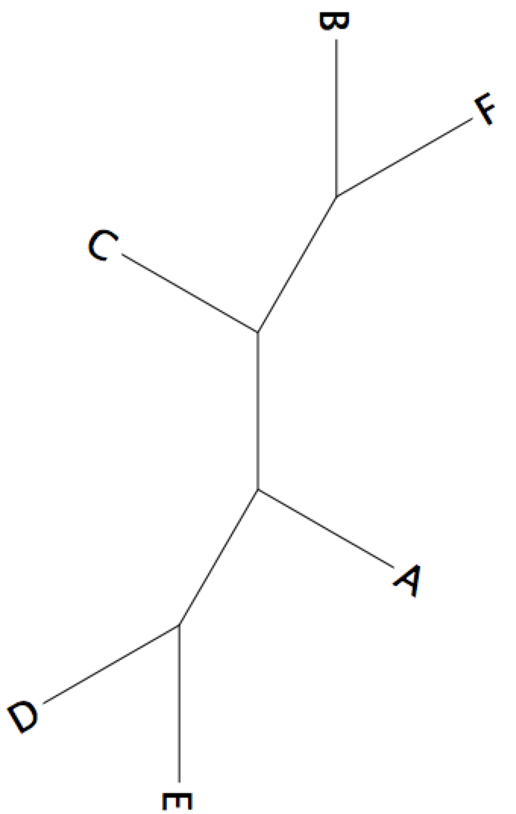

(A,(E,D)),(C,(B,F));

# Exercise 1 – Part 3.a




(A,(E,D)),(C,(B,F));

(A,(E,D),(C,(B,F)));



# Exercise 1 – Part 3.b

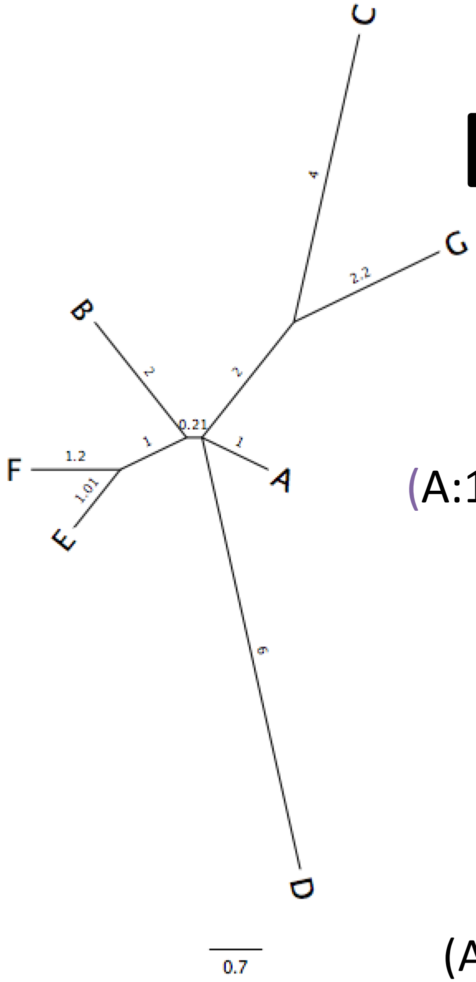
(A:1,D:6,([E:1.01,F:1.2]:1,B:2):0.21,(C:4,G:2.2):2):1);

# Exercise 1 – Part 3.b

(A:1,D:6,([E:1.01,F:1.2]:1,B:2):0.21,(C:4,G:2.2):2):1);

Use correct parentheses

(A:1,(D:6,((E:1.01,F:1.2):1,B:2):0.21),(C:4,G:2.2):2):1);



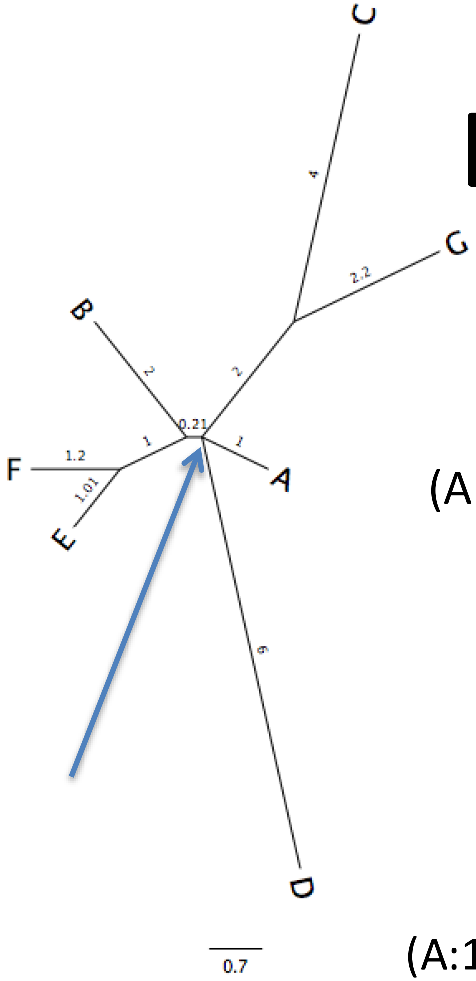

# Exercise 1 – Part 3.b

(A:1,D:6,((E:1.01,F:1.2):1,B:2):0.21,(C:4,G:2.2):2):1);

Remove incorrect branch length

(A:1,D:6,((E:1.01,F:1.2):1,B:2):0.21),(C:4,G:2.2):2);

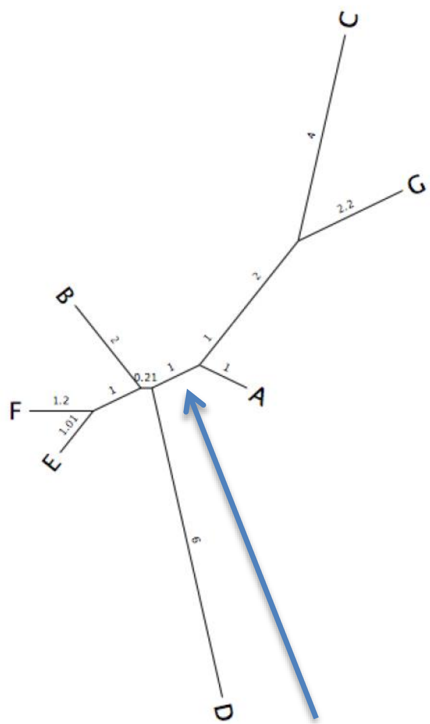
# Exercise 1 – Part 3.b




(A:1,D:6,((E:1.01,F:1.2):1,B:2):0.21,(C:4,G:2.2):2);

Resolve multi-branching point

(A:1,**(**D:6,((E:1.01,F:1.2):1,B:2):0.21**):1**,(C:4,G:2.2):2);




**Note**: There are many ways to solve the multifurcation, this is just one of the many.

# Exercise 2 – Overview

**Small theory recap**

Phylogenetic trees are used to represent the evolutionary relationships between a group of related sequences/species/genes. For tree construction, several computational methods are available. These are as follows:

1. **Distance Based Method**: Neighbor Joining, etc. These require a distance measure between the sequences.
2. **Maximum Parsimony**: The tree based on this method will provide the minimum number of evolutionary steps to produce the sequences.
3. **Maximum Likelihood**: This method uses an expected pattern of mutational changes from one DNA base to another with probability calculations to find the most likely arrangement of branches that generates the set of sequences.

$$L = p(\text{data} \mid \text{tree, branch lengths, model})$$

The ML algorithm searches different trees and branch lengths to find the $\mathbf{L_{max}}$. It works as following

**LOOP OVER**

**Generate tree topology** è **Optimize branch lengths** è **Retain if result improved**

# Exercise 2 – Part 1.d

**Can you identify which command line arguments are required (non-optional) for IQ-TREE to be executed?**

# Exercise 2 – Part 1.d

**Can you identify which command line arguments are required (non-optional) for IQ-TREE to be executed?**

**-s** name of the alignment file

# Exercise 2 – Part 1.d

**What are other basic parameters that are optional, but quite useful?**

# Exercise 2 – Part 1.d

**What are other basic parameters that are optional, but quite useful?**

**-m** model of binary (morphological), nucleotide, multi-state, or amino acid substitution

**--prefix** prefix used to name output files

**--seed** random seed for all stochastic steps

# Exercise 2 – Part 2

**Running IQ-TREE on the MFS-1 dataset**

iqtree

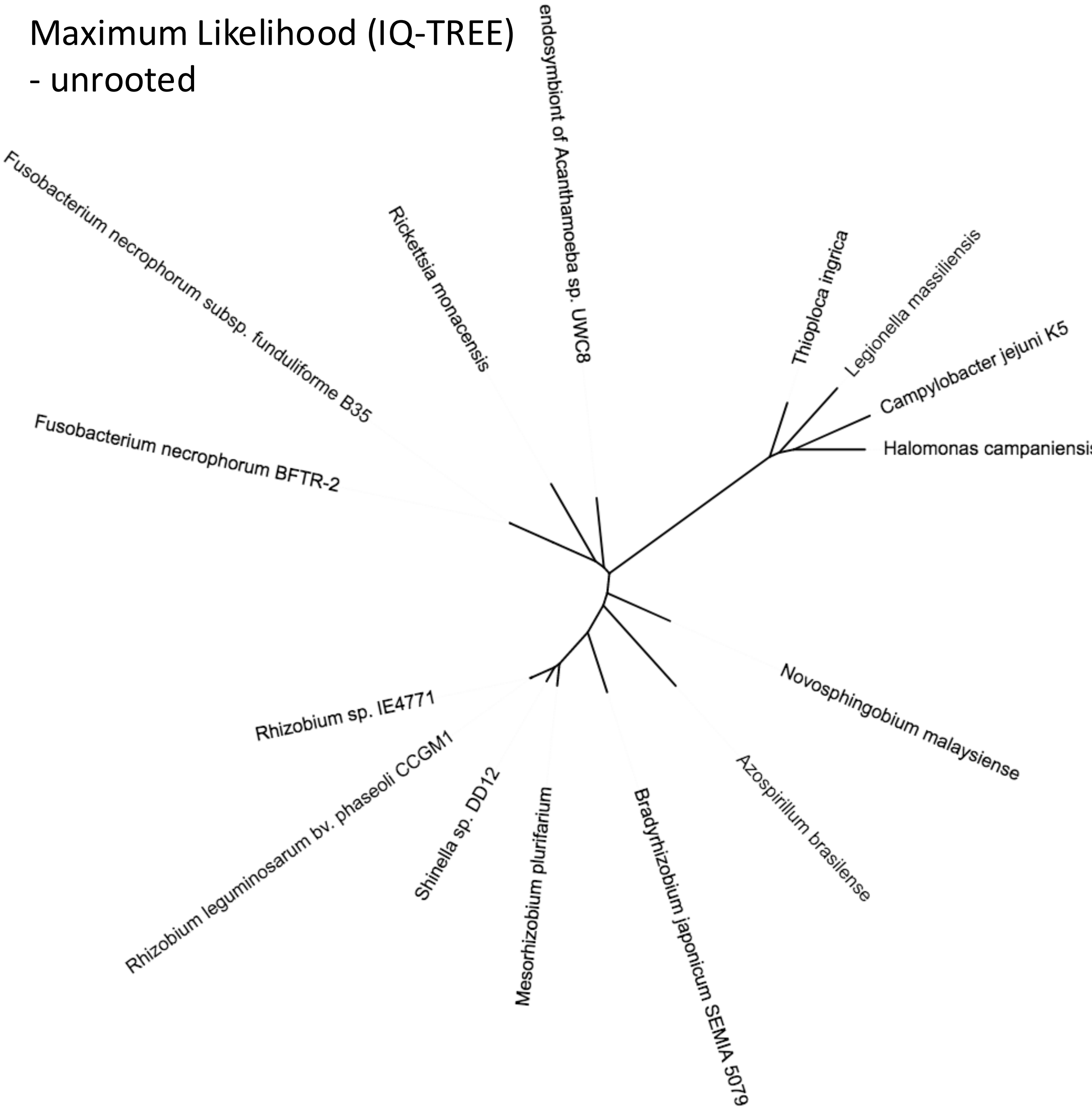
-s mfs_domain_proteins_aligned_taxnames.fa

-m MFP

--prefix mfs.auto

--seed 123

# Maximum Likelihood (IQ-TREE) - unrooted



**Fusobacterium**

---

- Pseudomonadati
  - Campylobacterota
    - Epsilonproteobacteria
      - Campylobacterales
        - Campylobacteraceae
          - **Campylobacter**
  - Pseudomonadota
    - Gammaproteobacteria
      - **Thioploca**
      - **Halomonas**
      - **Legionella**
    - Alphaproteobacteria
      - **Novosphingobium**
      - **Rickettsia**
      - Hyphomicrobiales
        - **Mesorhizobium**
        - Rhizobiaceae
          - **Shinella**
          - **Rhizobium**
        - **Bradyrhizobium**
      - **Azospirillum**

**Kingdom: Fusobacteriati**

- **Fusobacterium**

---

**Kingdom: Pseudomonadati**

- Pseudomonadati
  - Campylobacterota
    - Epsilonproteobacteria
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          - **Campylobacter**
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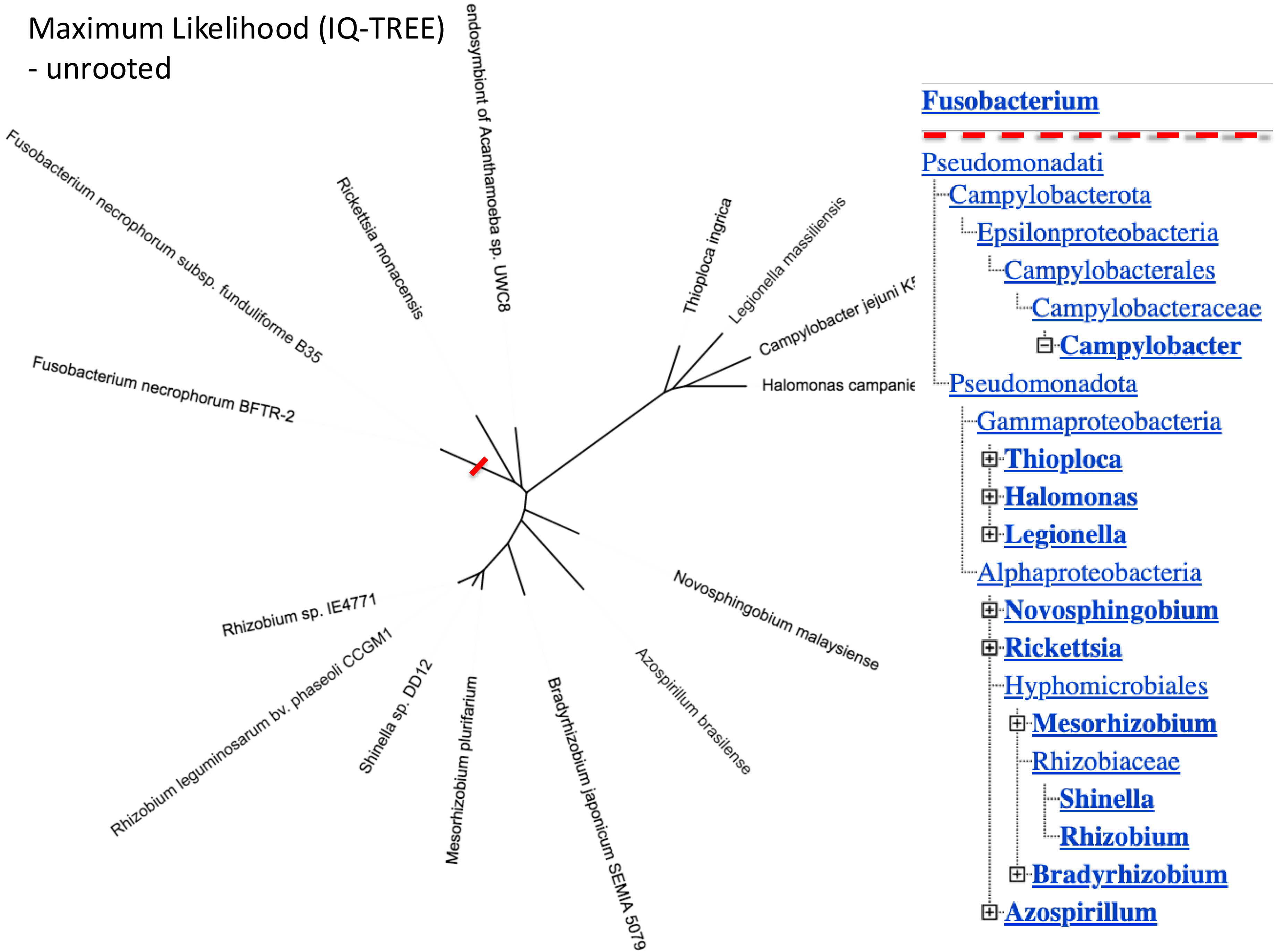
Maximum Likelihood (IQ-TREE) - unrooted



# Revising bacterial taxonomy

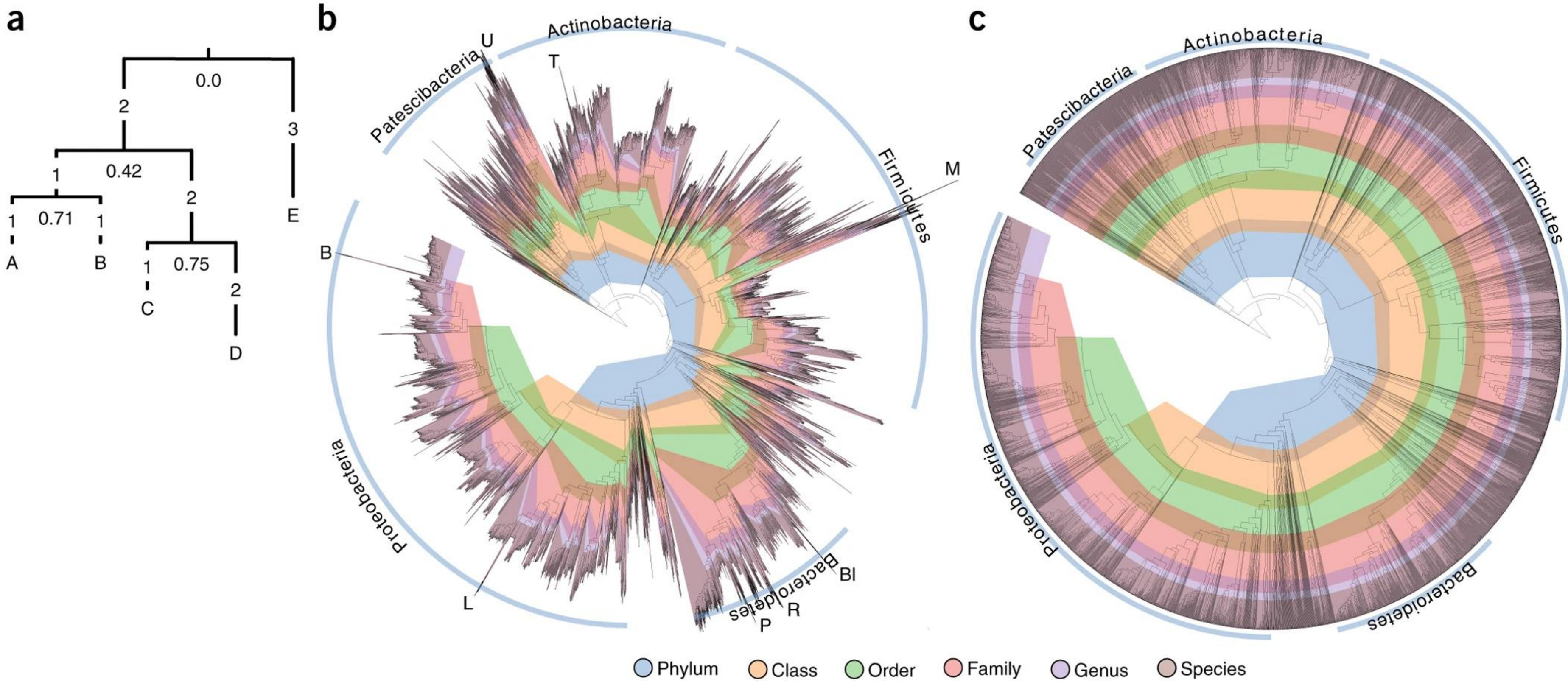
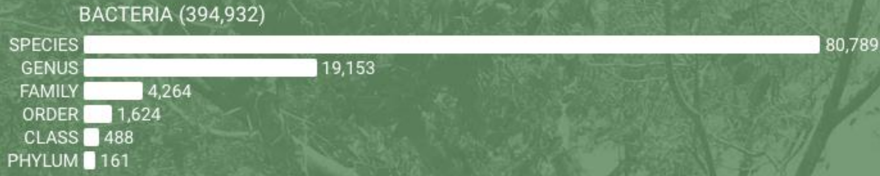



Image from: Parks DH et al., *Nat Biotechnol* **36**, 996–1004 (2018). https://doi.org/10.1038/nbt.4229




Australian Centre for Ecogenomics

Welcome to GTDB

# GENOME TAXONOMY DATABASE

402,709 genomes

Release 08-RS214 (28th April 2023)


THE UNIVERSITY OF QUEENSLAND AUSTRALIA

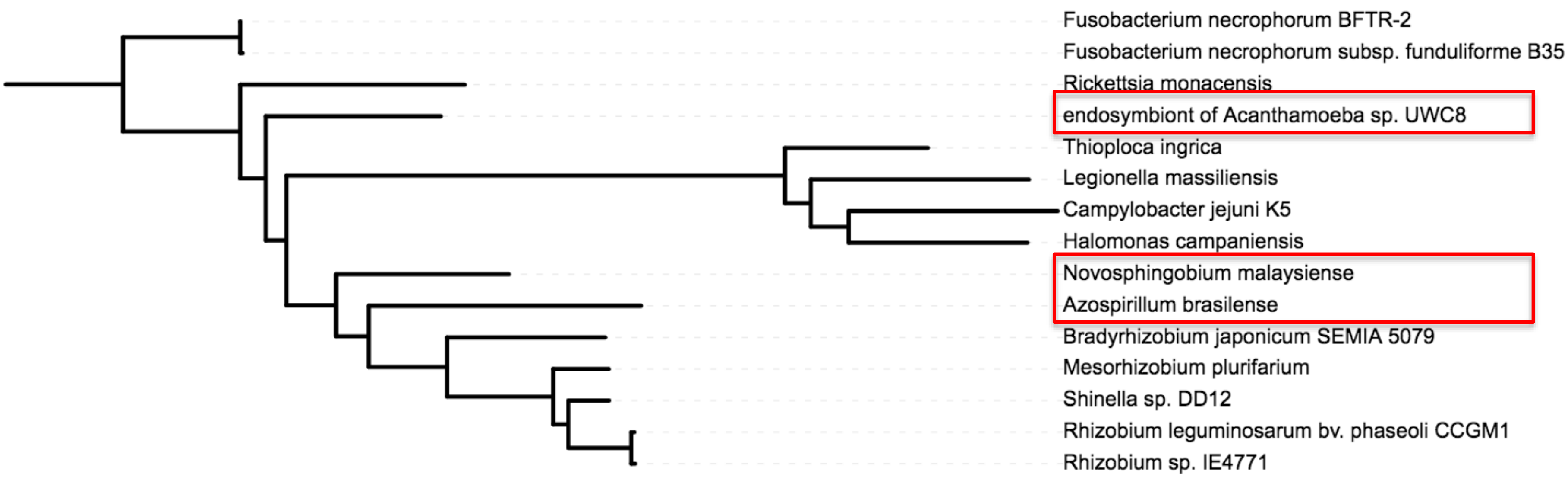
# Maximum Likelihood (IQ-TREE, rooted on Fusobacteria)

# UPGMA (rooted on Fusobacteria)


# Maximum Likelihood (IQ-TREE, rooted on Fusobacteria)

## UPGMA (rooted on Fusobacteria)


## Maximum Likelihood (IQ-TREE, rooted on Fusobacteria)



# Neighbor joining (rooted on Fusobacteria)


# Maximum Likelihood (IQ-TREE, rooted on Fusobacteria)

## Neighbor joining (rooted on Fusobacteria)


## Maximum Likelihood (IQ-TREE, rooted on Fusobacteria)

# Exercise 3 – Part 1

## **Sequence alignment**

```
HTLV            SHRFKNLGAQTGELWNTFLKTAAPLAPVKALMPVFTLSPVIINTAPCLFSDGSTSR----
HIV1B5          --------------WE-FVNT-PPLVKL-----WYQLEKEPIVGAETFYVDGAASRETKL
HIV1H2          --------------WE-FVNT-PPLVKL-----WYQLEKEPIVGAETFYVDGAANRETKL
HIV1PV          --------------WE-FVNT-PPLVKL-----WYQLEKEPIVGAETFYVDGAANRETRL
HIV1N5          --------------WE-FVNT-PPLVKL-----WYQLEKEPIIGAETFYVDGAANRETKL
HIV1OY          --------------WE-FVNT-PPLVKL-----WYQLEKDPIVGAETFYVDGAANRETKL
HIV1ND          --------------WE-FVNT-PPLVKL-----WYQLEKEPIIGAETFYVDGAANRETKL
HIV1Z2          --------------WE-FVNT-PPLVKL-----WYQLEKEPIIGAETFYVDGAANRETKL
HIV1MN          --------------WE-VVNT-PPLVKL-----WYQLEKEPIVGAETFYVDGAANRETKK
HIV1U4          --------------WE-FVNT-PPLVKL-----WYQLEKDPIAGAETFYVDGAANRETKL
SIVCZ           --------------WE-FINT-PPLVKL-----WYSLETEPIPTTDTYYVDGAANRETKT
HIV2CA          --------------WD-FVST-PPLVRL-----AFNLVGDPIPGTETFYTDGSCNRQSKE
HIV2RO          --------------WD-FVST-PPLVRL-----AFNLVGDPIPGAETFYTDGSCNRQSKE
HIV2SB          --------------WD-FVST-PPLVRL-----AFNLVKDPIPGAETFYTDGSCNRQSKE
HIV2KR          --------------WD-FVST-PPLVRL-----AFNLVKDPIPGEETFYTDGSCNRQSKE
HIV2ST          --------------WD-FIST-PPLVRL-----VFNLVKDPILGAETFYTDGSCNKQSRE
HIV2D1          --------------WD-FVST-PPLVRL-----TFNLVGDPIPGTETFYTDGSCNRQSKE
HIV2G1          --------------WD-FVST-PPLVRL-----TFNLVGDPIPGAETFYTDGSCNRQSKE
Smanga_S4       --------------WD-FVST-PPLVRL-----VFNLVKEPIQGAETFYVDGSCNRQSRE
Smanga_SP       --------------WD-FVST-PPLVRL-----VFNLVKEPIQGAETFYVDGSCNRQSRE
                              *: .:.* .**. :      : *    *      : **: .:
```

```
HTLV            ---------------------------------LANTGASR-------------------
HIV1B5          DRQ-GTVSFNFPQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGG
HIV1H2          DRQ-GTVSFNFPQVTLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGG
HIV1PV          DRQ-GTVSFNFPQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGG
HIV1N5          DRQ-GTVSFSFPQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMNLPGRWKPKMIGG
HIV1OY          DRQ-GTVSFNLPQITLWQRPIVTIKIGGQLKEALLDTGADDTVLEEMNLPGRWKPKMIGG
HIV1ND          ERQ-GTVSFSFPQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEINLPGKWKPKMIGG
HIV1Z2          ERQ-GTVSFNCPQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMNLPGKWKPKMIGG
HIV1MN          DRQ-GPVSFSFPQITLWQRPIVTIKIGGQLKEALLDTGADDTVLGEMNLPRRWKPKMIGG
HIV1U4          ERQ-GTDSFSFPQITLWQRPLVTVKIGGQLIEALLDTGADDTVLEDINLPGKWKPKIIGG
SIVCZ           -RE-QSISTNLPQITLWQRPLIPVKVEGQLCEALLDTGADDTVIERIQLQGLWKPKMIGG
HIV2CA          MQG-DNRGLAAPQFSLWKRPVVTAHIEGQPVEVLLDTGADDSIVAGIELGSNYSPKIVGG
HIV2RO          IQGATNRGLAAPQFSLWKRPVVTAYIEGQPVEVLLDTGADDSIVAGIELGNNYSPKIVGG
HIV2SB          TQR-DDRGLAAPQFSLWKRPVVTAYIEDQPVEVLLDTGADDSIVAGIELGSNYSPKIVGG
HIV2KR          TQR-GDRGFAAPQFSLWKRPVVTAYVEGQPVEVLLDTGADDSIVAGIELGSNYSPKIVGG
HIV2ST          MQR-DDRGLAAPQFSLWKRPVVTAHVEGQPVEVLLDTGADDSIVAGVELGSNYSPKIVGG
HIV2D1          PQR-GDRGLATPQFSLWKRPVVTAFIEDQPVEVLLDTGADDSIVAGIELGDNYTPKIVGG
HIV2G1          SQR-GDRGLAAPQFSLWKRPVVTAYIEVQPVEVLLDTGADDSIVAGIQLGDNYVPKIVGG
Smanga_S4       -QG-GDRGFAAPQFSLWRRPVVTAYIEEQPVEVLLDTGADDSIVAGIELGPNYTPKIVGG
Smanga_SP       -QG-GNGGFAAPQFSLWRRPIVTAYIEEQPVEVLLDTGADDSIVAGIELGPNYTPKIVGG
                                                 * :***.
```

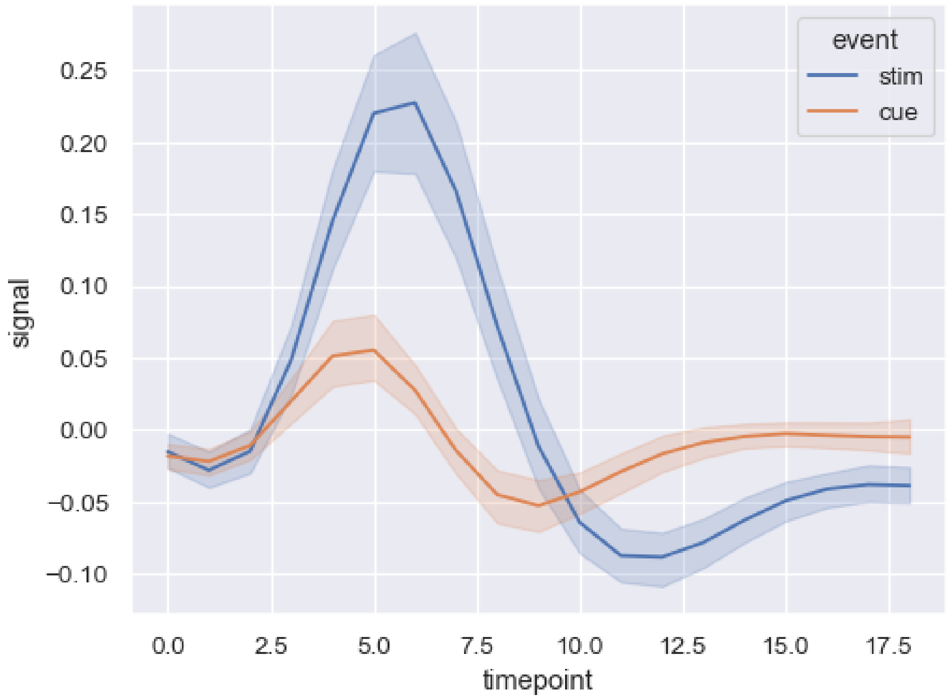
# Exercise 3 – Part 2

**What is bootstrapping?**

# Exercise 3 – Part 2

## What is bootstrapping?

**General statistical technique** that repeatedly resamples observed data to estimate the variability or confidence of a statistic (e.g. the mean or median)



# Exercise 3 – Part 2

## **What is bootstrapping?**

General statistical technique that repeatedly resamples observed data to estimate the variability or confidence of a statistic (e.g. the mean or median)

For phylogenetics, the bootstrapped statistic is **tree topology** (or clade support).

# Exercise 3 – Part 2

## What is bootstrapping?

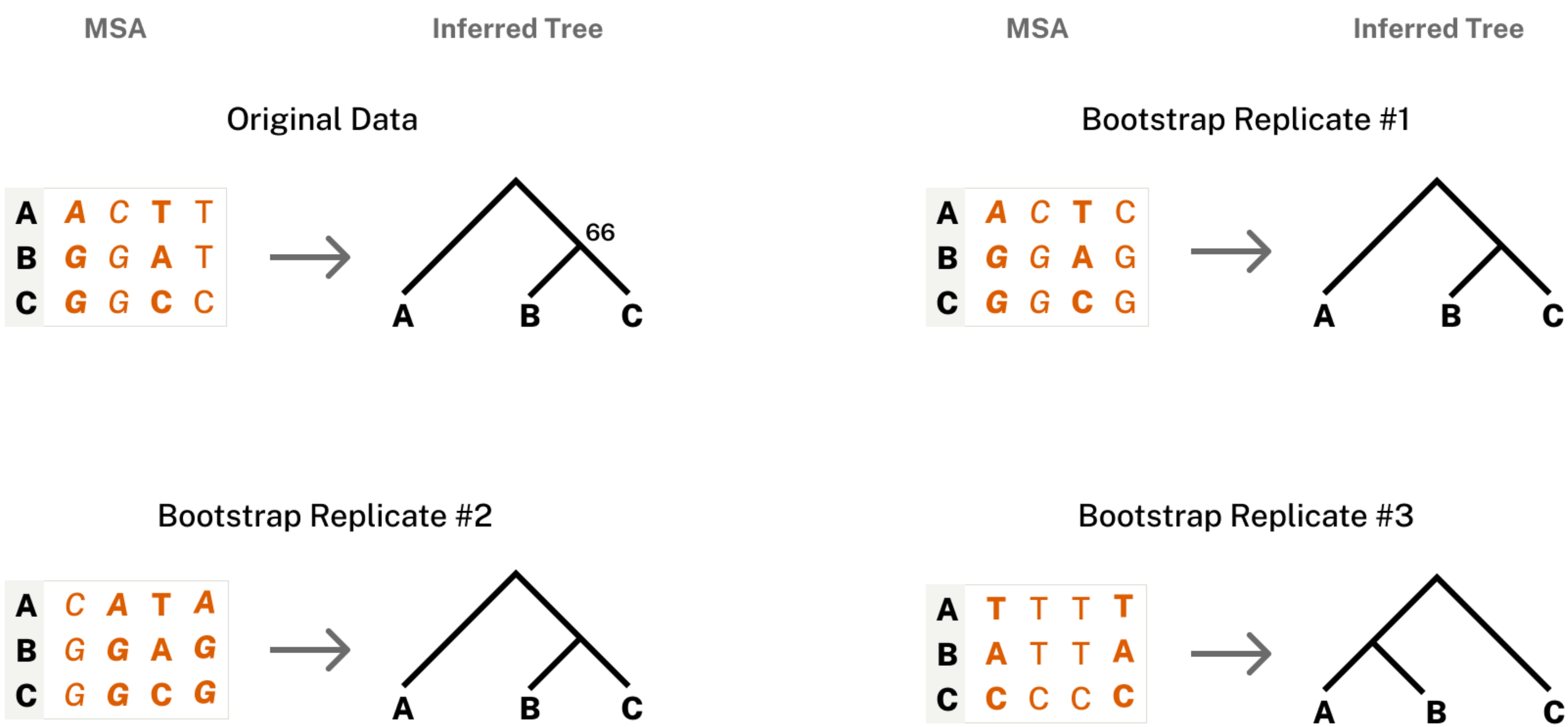



Image from: https://www.geneious.com/guides/understanding-phylogenetics

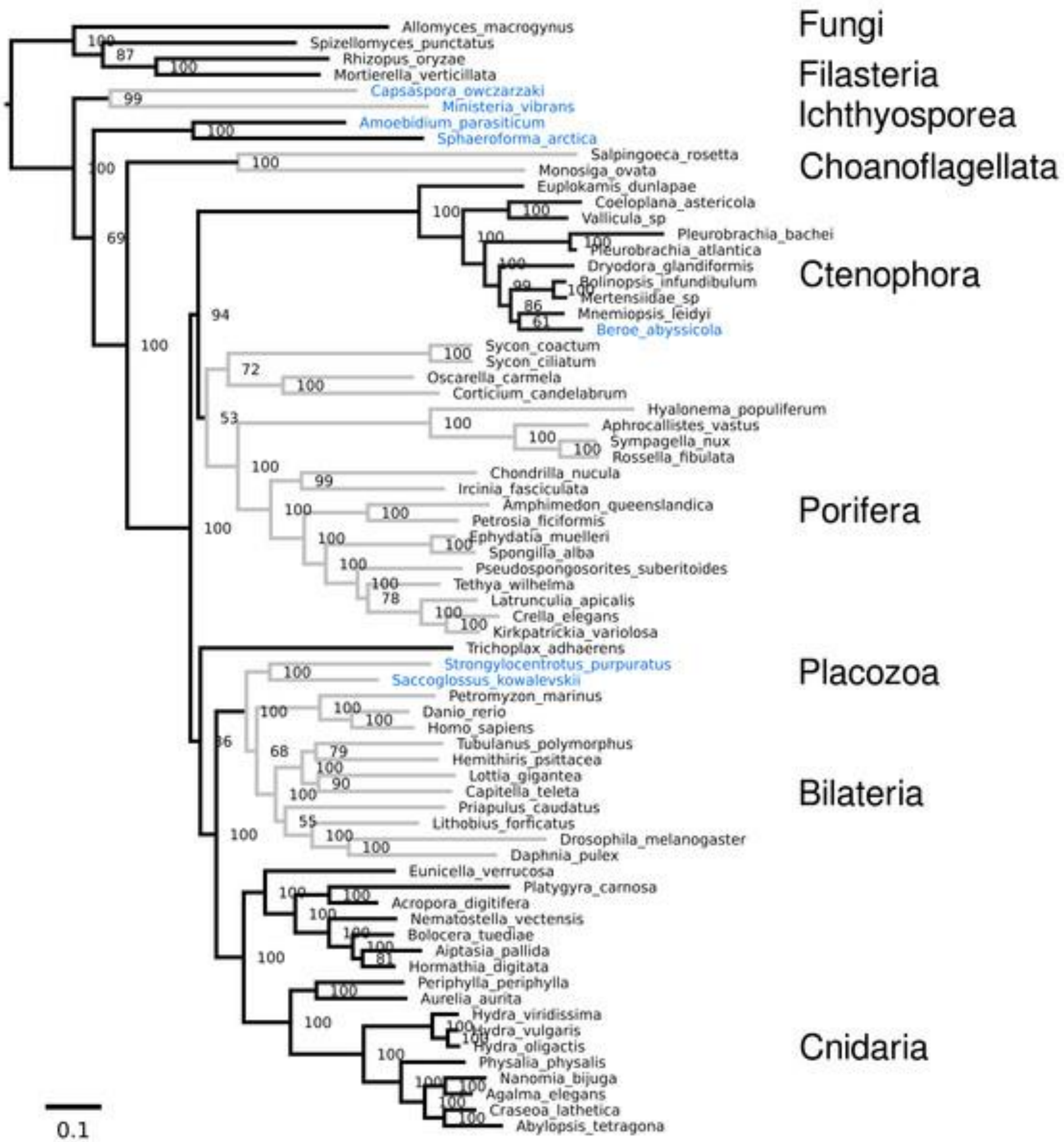



Image from: Warren R et al., *PeerJ*, **8** (2020)

# Exercise 3 – Part 2

**What are limitations of bootstrapping?**

# Exercise 3 – Part 2

## What are limitations of bootstrapping?

**Bootstrap values are not probabilities** of a clade (or tree) being correct – they measure consistency, not truth.

They only quantify stability **with respect to the dataset** - high support suggests robustness (not necessarily correctness), low support suggest instability (not necessarily incorrectness).

**However:** given good data (!) and appropriate methods, they can provide reasonable support for evolutionary hypotheses.

# Exercise 3 – Part 2

**What do the following more advanced parameters do?**

-b / -B

# Exercise 3 – Part 2

**What do the following, slightly more advanced parameters do?**

**-b** standard bootstrap procedure

**-B** ultra-fast approximate bootstrap procedure (UFBoot)

- orders of magnitude faster
- non-standard interpretation: UFBoot FAQ
- can be further supported with approx. likelihood-ratio tests (-alrt)

# Exercise 3 – Part 2

**Why did you get a smaller number of bootstraps than specified?**

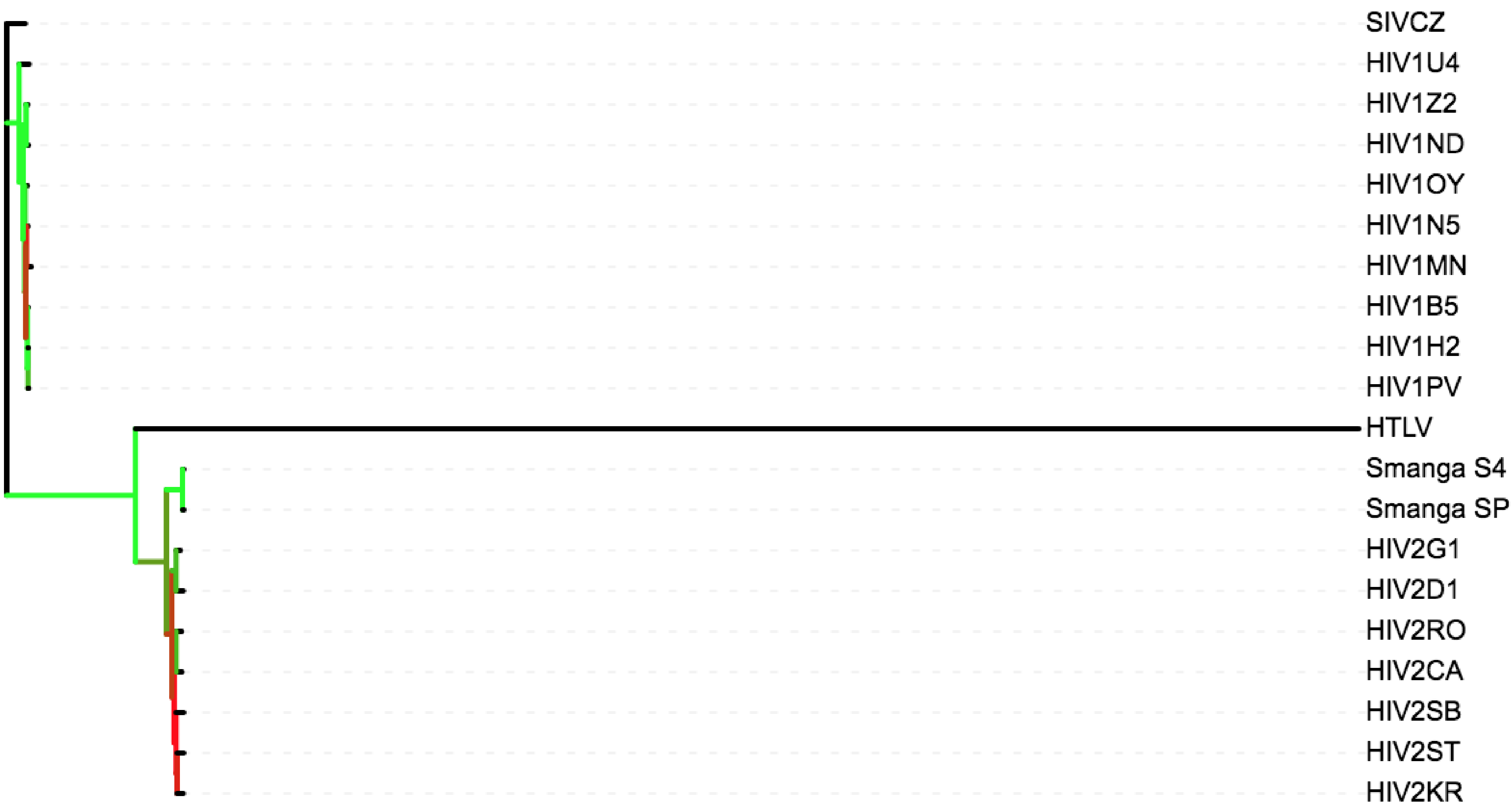
# Exercise 3 – Part 2

**Why did you get a smaller number of bootstraps than specified?**

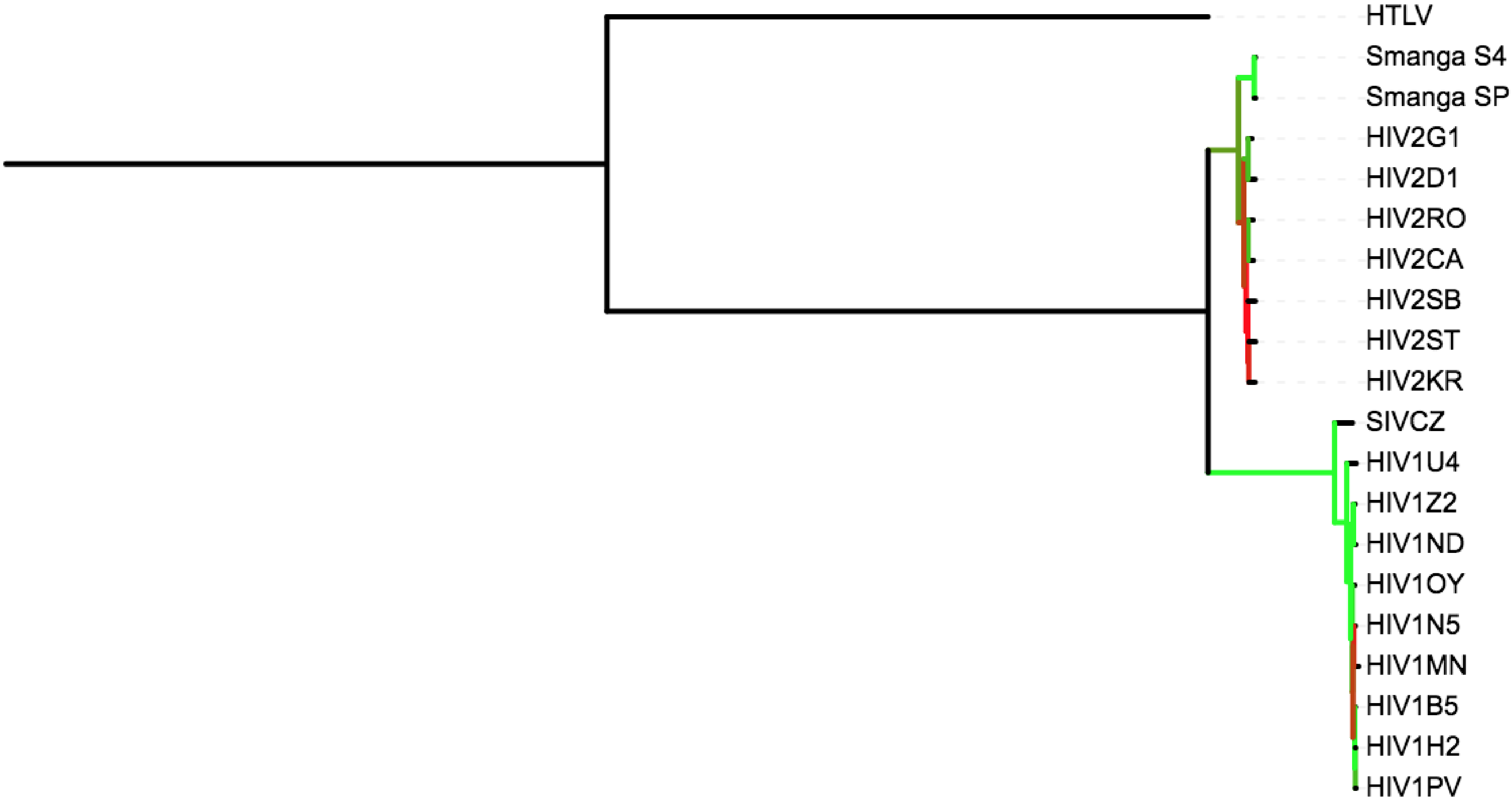
UFBoot uses a “stopping rule” heuristic to avoid generation of unnecessary bootstrap trees

- when split frequencies have converged (do not “substantially” change anymore), stop the algorithm

# Exercise 3 – Part 3



# Exercise 3 – Part 4



# Exercise 3 – Part 4